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Mental representation without neural representation

Understanding the evidence

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Abstract

It is common to hear neural states and processes described in representational terms, where it is alleged that neurons function to represent stimuli, categories, motor commands and various other things. Evidence for such a role is typically based upon the ways neurons reliably respond to specific stimuli. In a recent paper, Pohl et al. (2024) have offered a more formalized account of neural representation evidence that utilizes Shannon's information theory. Despite the ingenuity of their theory, we argue that Pohl et al. fail to present a successful account of evidence for neural representation, and indeed, repeat common mistakes that other researchers make in supporting neural representations. We offer a more in-depth and rigorous analysis of such evidence, and argue for a more skeptical conclusion: What commonly gets presented as evidence for neural representations actually isn't. While we support the reality of cognitive representations at higher levels of analysis, we do not believe compelling evidence has been presented that supports assigning a representational role for neural states and structures.

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1 Introduction

Mainstream cognitive science is overwhelmingly committed to what is commonly referred to as the "representational theory of mind" – the thesis that to understand cognition, we must posit the existence of internal cognitive states that serve to represent a variety of entities, properties, events, states of affairs, etc. (see, for example, Fodor, 1975; Neander, 2017; Ramsey, 2007; Shea, 2018). This is unsurprising given the degree to which commonsense psychology has influenced cognitive science and is itself committed to representational states, such as thoughts, memories, perceptual sensations and imaginings. However, there is less of a consensus on exactly how and where representational states belong in our scientific theorizing. Indeed, the very concept of representation as it pertains to cognition is extremely imprecise, variegated and ill-defined (Favela and Machery, 2023). While researchers seem convinced that representation is central to our theorizing, they seem less sure about how and why this is so.

In this essay, our main goal is to challenge a very common area where representations are increasingly claimed to play a pivotal role in the mind sciences. Neural states, processes and mechanisms, as such, are regularly depicted as representational in nature. And yet it is far from clear that there is compelling evidence in support of the idea that neural states are actually playing any sort of representational role or are functioning in a manner sufficiently similar to representational entities we are familiar with. While we allow that it may be *possible* for investigations to present compelling reasons for treating neural data as representations, we do not think this has yet been accomplished, and have a hard time seeing how it could be accomplished. Instead, we believe researchers and writers have a habit of assuming a representational role for neural activity, and then go on to describe a variety of quite possibly non-representational activities using representational language. In the past, each of us has tried to expose this erroneous habit in other works (for example, Krakauer, 2022; Ramsey, 2007). And yet, what we regard as a sort of "neural representation attribution error" has, if anything, only grown in popularity. Our goal is to try to, once again, point out the flaws in this reasoning.

Anti-representational concerns in the past have often focused upon the roles attributed to alleged representational states, and then questioned whether such roles (e.g., serving as a causal mediator) were sufficiently "representation-like" in nature. This was the idea behind questioning whether or not neural states met what one of us (Ramsey, 2007) has called the "job-description challenge". In this essay, while we intend to continue this focus upon representational roles, our approach will be somewhat different. We want to focus upon a question that all researchers should care about regardless of their stance on the reality of neural representations: just what, exactly, should or should not count as evidence for the proposition that neural states or structures are playing a representational role? While these are closely related concerns, the metaphysical task of specifying criteria sufficient for a representational role is not the same as the epistemological task specifying what might count as good evidence *for* such a role being played. Until very recently, there has been little direct examination of just how and why various findings are supposed to provide evidence for the representational hypothesis. Thankfully, Pohl et. al. (2024) have quite recently offered a framework for addressing this question. However, as we will argue, their analysis suffers from serious flaws that reveal the problematic nature of how representations are invoked and defended at the neural level. We hope to bring these flaws to your attention.

To do this, our discussion will have the following organization. In the next section we raise a number of important preliminary issues regarding evidence for neural representations that need to be addressed to set the stage for our analysis. Next, we will focus upon the analysis of evidence for neural representations offered by Pohl et. al. to evaluate its's plausibility. Our claim will be that while Pohl et. al. accurately capture a general attitude regarding neural representation evidence, that attitude is not justified given various facts about evidence and representation. We will also extend this critique to some other accounts in the literature, and discuss common assumptions about alleged evidence for neural representation, based upon factors like how the neural state is triggered, or how neural activity clusters in certain ways. We will offer our critical assessment of these accounts and try to explain why the focus should be upon efferent, rather than afferent

conditions. Finally, we'll offer some considerations regarding what *could* qualify as evidence of something in the brain playing a representational role, and note that while these considerations support the invoking of representational states at higher levels of analysis, they cast doubt on the prospects of *neural* representations.

2 Preliminary issues

2.1 Four notions of cognitive representation

First, it will be helpful to clarify the central target of our inquiry. The explanatory status of cognitive representations within cognitive systems is complicated and tricky. On the one hand, representational states like memories or imagery are among the explanada of cognitive science – they are among the phenomena we are trying to explain. And yet, it is also generally assumed that we must invoke representational states as explanantia – as posits that help explain the cognitive phenomena (for example, explaining how someone buys appropriate furniture in a store requires positing an internal representation of the rooms in your home). One reasonable and popular assumption is that representations actually occupy both roles in modern cognitive science – they are both explanatory targets and explanatory postulates. A further issues concerns the level at which representational activity should be described. Should we think of representational cognition as occurring at a fairly abstract or computational level, as features of a person or full-blown mind (as *mental* representations)? Or should we think of representations as occurring at the "sub-personal" level of analysis, perhaps even as basic neuronal states and processes (as *neural* representations)? Once again, a popular outlook is that both levels of description are suitable for invoking representations.

To capture these distinctions, we can use a 2×2 matrix with one axis involving representations as either explananda or explanantia, and another axis as dividing representations into either personal (i.e., mental) or as sub-personal (i.e., neural) levels of analysis, as shown in Figure 1.

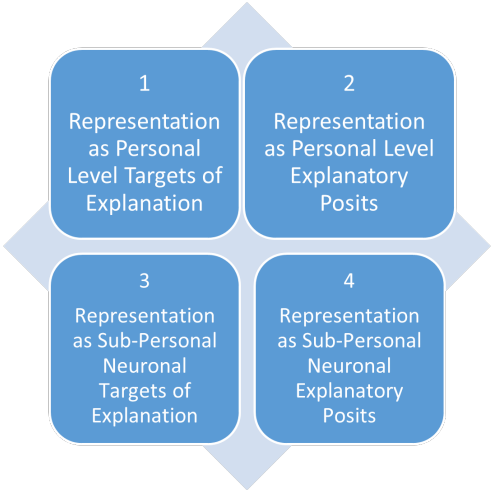

Figure 1: Four notions of cognitive representation.

We can thereby recognize four ways in which representations can be invoked: 1) As a personal level phenomena to be explained (e.g., perceptions, thoughts, experiences or memories); 2) As personal/psychological-level posits that explain some form of cognition or behavior (e.g., working memory or concepts); 3) At the sub-personal level as a property of neurons that needs to be explained (e.g., place cells) and 4) At the sub-personal level as a capacity of certain neural states that explains cognitive behaviors (.e.g., hippocampal replay as the basis for navigational planning). With this framework in mind, we can clarify that our aim is to explain our skepticism about the third and fourth ways in which representations are invoked – as neural representations we are trying to understand, and as neural representations invoked to explain a variety of different capacities. We are thus not challenging the invoking of representational states at the personal level of analysis, either as targets of investigation and theorizing, or as explanatory posits; we are not interested in promoting a sweeping version of anti-representationalism associated with, say, the radical enactivist perspective (Hutto and Myin, 2013). Indeed, we are generally sympathetic

with the perspective that higher-level operations in cognitive processes operate on representational states. We want to emphasize that we are ***not*** arguing against the reality of, or evidence for, mental representations. Nor are we are arguing against the reality of representational behavior. Our focus is on the question of whether or not there is evidence to suggest that *neural* states and activities, as such, should be treated as representational in nature.

We should also make clear that we intend to use the term 'representation' to designate states or structures that, well, actually represent – that have the task of carrying or relaying information, that have intentional content (i.e., are about something), that have the capacity to misrepresent and be wrong, and that serve to model or indicate other conditions. This is the standard notion of representation at the heart of cognitive science and that has a long history in our attempts to make sense of the mind. However, we are aware that there may be some who employ the term to designate nothing more than a state that functions as, say, a causal mediator or as something that simply responds to specific conditions. We believe that these deflationary definitions are wrongheaded and severely undermine scientific understanding by creating conceptual confusion, miscommunication, and by suggesting there is nothing special about representational systems. It is akin to claiming that plants possess consciousness by simply defining consciousness as the capacity to react to stimuli. Valuable critical analysis and meaningful theoretical dialogue require a shared language; consequently, the standard meaning of critical terms should be respected and not significantly altered without good reason.

2.2 The epistemology of evidence

Following standard practice, we regard evidence as any observation, finding, datum or fact that provides support for a given hypothesis (Kim, 1993). We recognize that the quality of evidence can come in degrees, depending upon how much support (or the strength of justification) it provides for the hypothesis in question. We also recognize that the quality of the evidence can be heavily context-dependent, such that an observational finding in one context can provide compelling evidence for a hypothesis, although the same evidence would only provide very weak evidence in a different context.

For our purposes, a critical contextual factor that looms large in our discussion will be how competing hypotheses impact the evidential value of a finding. Generally, the greater the number of alternative hypotheses that are compatible with the finding in question, the poorer that finding is for any particular hypothesis. For example, a distinctive footprint might function as strong evidence for the hypothesis that a certain suspect – say, Smith – was at the crime scene because Smith wears the type of shoe that exactly matches the footprint. However, the value of this evidence depends upon whether (and the degree to which) that tread is unique to Smith's shoes. If a number of other equally likely suspects (Jones, Brown, Johnson and Miller) wear the exact same type and size of shoes, then although the footprint could still qualify as evidence that Smith was at the scene, its evidential status for that hypothesis would be substantially weakened. The greater the number of alternative, equally likely suspects there are that could have created the exact same footprint, the less we should treat the footprint as pointing to Smith as the culprit.

A slightly different articulation of this relationship between evidence and hypothesis has been called the "Prime Principle of Confirmation" (Collins, 2003). This is the view that evidence E counts as good evidence of hypothesis H1 over hypothesis H2 if E is more likely given H1 than given H2. In our example, the footprint would support the hypothesis Smith alone was present over the hypothesis that Jones was present only if the footprint fits Smith's shoes and not Jones'. If the footprint is no more likely given the Smith hypothesis than the Jones hypothesis, then it would fail as evidence for the proposition that Smith alone was present over the proposition that Jones was the culprit.

One further related outlook on evidence is that a confirming evidential finding is best explained by the hypothesis it confirms (Douven, 2021). Abductive inference involves inferring a hypothesis is correct when it provides the superior explanation for an observation. In our example, if Smith alone wears the relevant shoe, the hypothesis that Smith was at the

crime scene does provide a superior explanation for the footprint, and thus the footprint confirms that hypothesis. But if various other suspects share the exact same sort of shoe with Smith, then the hypothesis that Smith alone was present would fail to provide the best explanation of the finding.

Overall, then, we can say that purported evidence E serves as confirming evidence for hypothesis H when E is uniquely compatible with H and not an array of other equally likely hypotheses, E is more likely given H than any other hypothesis, and if E is best explained by H in contrast to other hypotheses. Moreover, these criteria can be applied in our assessment of the evidence for neural representations. If a finding uniquely supports the hypothesis that neural states or structures are playing a representational role as opposed to a host of other equally plausible roles, is more likely given the representational hypothesis, and if the representational hypothesis best explains the finding in question, then such a finding should be seen as compelling evidence for neural representations. On the other hand, if a finding is (as far as we can tell) equally compatible with a number of plausible alternative, non-representational roles for the neural states or structures in question, or if the finding is equally likely given alternative, non-representational roles, or if the finding appears to be explained just as well by other non-representational hypotheses, then such a finding should not be treated as persuasive evidence for the claim that the state or structure playing a representational role, as opposed to doing something else. As a preview, our skepticism about claims of neural representation is driven largely by the fact that the purported observations are actually compatible with, just as likely under, and appear to be equally well explained by other non-representational hypotheses.

2.3 Evidence for exactly what?

A complicating factor in our evaluation of representational evidence is the fact that the specific modality of the relevant hypothesis is somewhat unclear. Sometimes the pertinent hypothesis is treated as something like, given what we have found, we *must* regard certain neural states as representations. Thus, to count as compelling evidence, the finding would need to support the claim that a representational interpretation of neuronal neural activity is practically unavoidable. However, this strikes us as too strong, since no particular functional analysis of any neural system is absolutely necessary. It is always in principle possible to describe neuronal activity in purely physiological, causal/physical terms without invoking any specific functional roles or higher structures. Going the other way, the relevant hypothesis might be framed as the proposition that is it *possible* to regard neuronal states as representations. However, this is far too weak. As Dennett (1978) pointed out, by adopting the "intentional stance", one can in principle describe practically anything in representational terms – even a rock sitting on the ground can be characterized as representing the proposition that it should sit very still.

The appropriate framing of the relevant hypothesis should be one that focuses upon the theoretical or explanatory value of invoking representational interpretation (Ramsey, 2007, Shea, 2018). The question we should ask is something like this: does the evidential finding support the invoking of a representational role for neural states in a manner that is explanatorily beneficial? Do the findings suggest that we ought to view these neuronal states as representations, such that if we fail to do so we would be missing out on properly understanding the mechanics of cognition? Thus, when we are looking for evidence of neural representations, we are looking for evidence that neural states are playing a representational role such that the representational interpretation of what the neurons are doing has a clear explanatory pay-off.

This point hints at one potential source of evidence for neural representation; namely, the explanatory and predictive success of cognitive models and theories that invoke neural representation as explanatory posits (Shea, 2018). If a theory that puts forward some explanatory posit is successful, then that provides at least indirect evidence that such posits are real. However, there are a couple of difficulties with this kind of evidential support in general, and specifically with regard to neural representations. Generally, the history of science reveals that theories can enjoy enormous explanatory and predictive success, despite being wrong and positing theoretical entities that don't exist or that are badly misconstrued. Scientists are

human, and are not immune to such biases as discounting disconfirming findings as irrelevant or exaggerating the theory's overall achievements. To count as indirect evidence, explanatory and predictive success needs to be well established, and even then not all aspects of a theoretical framework are guaranteed to be vindicated. Moreover, in the specific case of neural representation, further questions arise regarding the explanatory role of the posits, *qua*-representational states. If researchers simply assume or stipulate a representational role for posits without adequate justification, then one can have a successful theory that invokes representations in name only. We must always ask whether a non-representational interpretation of the internal state could work equally well in the theory or model that enjoys explanatory and predictive success.

2.4 Implementing representations vs. being representational

We believe that in some sense, at some level of analysis, neurons clearly instantiate mental representations. After all, we think cognitive activity in the brain involves personal level mental representations, and the brain is made up of neurons. Thus, one might suppose that we should fully embrace the reality of neural representations. But to think this would be to commit a version of the mereological fallacy – the faulty inference that because a global system has certain properties, the component elements must themselves possess such properties. In reality, a central tenet of mechanistic explanation is that the special states and properties that are implemented by the coordinated activity of component elements rarely apply to those component elements themselves. The cells in our body that contribute to the respiratory process are not themselves treated as breathing. Similarly, the fact that neural states and processes in some way give rise to representational cognition should never motivate us to suppose that the neural states themselves, as such, need to be characterized as demonstrable representations, at least at the level of analysis suitable for understanding the properties and activities of neurons. We already understand this for a variety of dimensions of the mind, such as the capacity to have conscious experience or the ability to learn a new language. Practically no one thinks that we should explain these cognitive processes by supposing neurons themselves are, as such, conscious or capable of learning language. Whatever evidential motivation there can be for treating neural states and structures as representations, it should not be that they are the neural implementers of personal-level mental representations.

This raises the thorny question of what should count as a neural representation as opposed to a higher-level, non-neural representation. What, exactly, is the demarcation criterion for identifying neural representations? While this a huge topic, one pertinent observation is that when researchers invoke neural representations, they typically invoke states whose representational capacity arises out of the conventional activities of neurons. A good example would be place cells in the hippocampus. Here, the correlation of firing rates of certain specific neurons in the hippocampus of a rat to specific locations (such as in a maze) is directly linked to the neurons' alleged role as representations. That is, something neurons are generally known to do – fire at different rates – is thought to be explanatorily salient to their role as representations. By contrast, with algorithmic or computational representations, the physical details of the implementing medium are, for the most part, irrelevant for a vehicle's status as a representation (Marr, 1982). Representation is thought to occur not because of the physical nature of the implementing medium, but instead because of, for example, their role in more abstract computations. Roughly, if something's status as a neural state has very little to do with its status as a representation, or if the invoking of a representation has nothing to do with the distinctive activity of neurons and neural states, and involve theories pitched at an explanatory level more abstract than neuroscience, then we find it difficult to see why the label "neural representation" would be appropriate.

It bears repeating the traditional "special sciences" perspective that helped serve as part of the foundation of cognitive science so many years ago (Fodor, 1974; Marr, 1982). With this perspective, it is recognized that corresponding to different levels of organization in the brain are different levels of theorizing and different explanatory frameworks. Emergence is a real feature of complex systems, and the kind of explanatory posits and

machinery that occur at higher levels of organization should not be invoked as explanatory posits or targets at more fundamental levels of organization. Computational representational models of things like hurricanes and markets are very real, but it would be a mistake to look for them among the circuits and wiring of an actual computer that implements such a model. So too, the reality of mental models in the brain should not lead us to assume that they exist and should be investigated at the level of neural structures and operations.

Does this mean that neuroscience is irrelevant to our understanding of mental representation? Of course not, just as cellular biology is not irrelevant to respiration or locomotion. Neuroscience is critical for our understanding how the sort of computational systems that employ cognitive representations are implemented in the brain. But as was pointed out long ago, we should not expect to find the same explanatory posits at lower levels of analysis (corresponding to lower levels of organization) that we find at higher levels of analysis and explanation (corresponding to higher levels of organization).

It might be protested that recent developments in cognitive neuroscience have repudiated the older outlook that psychology and neuroscience operate at distinct levels of explanation, refuting the view that there are conceptual and epistemological problems with trying to understand psychological phenomena at the level of neurons (see, for example, Churchland, 1986). In fact, many advocates of cognitive neuroscience believe that work on neural representation reveals how wrongheaded the old outlook separating explanatory tasks was (Thomson and Piccinini, 2018). They suggest, at least tacitly, that neural representations offer a kind of existence proof of how mental phenomena can be understood at the level of neural states.

Our perspective also stems from an examination of neural representation, but it reaches the opposite conclusion. The pitfalls and problems associated with neural representation (that we will discuss below) reveal precisely why the traditional division of labor still makes sense, and why it is often a mistake to try to capture certain psychological phenomena at the level of neuroscience. For now, the critical point is this: Evidential support for the existence of neural representations cannot come from the simple fact that mental representations are implemented in some form in the brain.

2.5 Why representational evidence is so elusive

A central problem that arises in any search for evidence of neural representations concerns the concept of representation itself. Our ordinary conception of representation involves something functioning as a content-bearing informer or stand-in. Our conception of inner cognitive representation is at least partially rooted in our conception of non-cognitive representation – things like gas gauges or road maps, or perhaps natural signs (like smoke used to indicate the existence of a fire). The needle on a gas gauge plays a representational role because it informs us of how much gas is left in the tank. It relays to our minds certain facts pertaining to matters beyond itself. The lines on a map represent roads because we treat the lines as surrogative stand-ins for roads. This may seem innocent enough, but something's functioning as an informer or surrogate in this way involves factors that are not easily identified as purely physical or causal in nature. Functioning as a representation is not like functioning as a pump or a lever, where we know exactly what sorts of causal relations and properties to look for that manifest such roles. Representation functionality suggests the conveyance of information to an intelligent mind capable of becoming informed by such things. Thus, any evidence that neurons are representing something would need to point to activity that somehow, in some way, involves a role sufficiently close to this ordinary notion of representation. This is true even for "technical" scientific conceptions of representation. If something is characterized as a representational role that has no similarity to our ordinary, commonsense notion of what a representation does, then such misattributions should be seen as simply that – a flawed mislabeling of reality.[1] The

[1] Imagine someone claiming that neurons function as wrenches. When it is pointed out that neurons do not play a functional role sufficiently similar to the role wrenches play, it would not suffice to reply that this a technical sense of 'wrench'. Any assignment of a role to neurons requires they do something sufficiently close to that role. Science is not a license to make up entirely new meanings of words.

difficulty, of course, is that there is no full-blown, "homuncular" inner mind treating the neurons as indicators or surrogates of anything (Dennett, 1978). At the same time, we know that complex visual hallucinations can happen in patients with various neurological conditions, with patients describing what they are "seeing". These visual experiences seem to be not unlike the description of an external portrait or picture. Exactly what is going on is not clear, but any evidence of neural representations will need to be accompanied by a better understanding of the sort of observable properties and relations that can implement such genuine representational activity.

3 Evidential desiderata of neural representation

Our focus is on the question of what does and does not qualify as good evidence for neural representations. As noted in the Introduction, we are fortunate in that this very topic has been recently addressed in detail by Pohl, et. al.. In their essay entitled "Desiderata of Evidence for Representation in Neuroscience" (2024), these authors make explicit what has largely been implicit with regard to the evidentiary role of various neural properties and relations in support of their having representational status. Thus, their account presents an excellent basis for our own analysis and concerns, as it serves as a good exemplar of more popular views.

Pohl et. al. stipulate 4 properties of neuronal states that, in their mind, can and should serve as evidence that such states function to represent some feature of the world. These are: 1) sensitivity to the feature in question (the neurons' activity is correlated in some way with the feature, and thus carries information about the feature); 2) specificity in that sensitivity (the neural state's different parameters are best explained by changes in the feature); 3) invariance in the responsiveness (the neural state's activity is not influenced by other variables or stimuli); and 4) functionality (the information carried by the neural state is in some way relevant to downstream, efferent processing). Following many others (see for example, Sayre, 1986), the authors employ Shannon's informational theory framework to provide a more mathematically robust calculus for measuring the information carried by neural responses to stimuli (Shannon, 1948). Their aim is to provide a more general formal framework for understanding and discussing neural representations. To illustrate how all this would work, they show how such a framework applies to a range of alleged neuronal representation systems such as V1 neurons responding to the orientation of stimuli, or neuronal activity in the Fusiform Face Area in response to facial stimuli.

Despite the ingenuity of this account, there are also a number of prima facie difficulties with it. For starters, the title is somewhat misleading as the authors do not actually present desiderata for evidence of representations. That is, they do not present desirable features that a finding itself should possess to qualify as legitimate evidence of neural representations. Instead, they simply present findings that they and others think qualify as evidence of representations. Desiderata of the evidence would be something more like what was presented above in Section 2.2, where we discussed some features that the alleged evidence would need to display to count as good evidence. Furthermore, a strict reading of the specificity and invariance criteria could make their presence difficult to determine. After all, no one can know the actual exclusive set of features and conditions neurons respond to (because no one has tested neural responses to every possible stimuli); moreover, if neurons do indeed respond to all and only the conditions they represent, then that would seem to raise difficulties for the possibility of misrepresentation, at least without further stipulated conditions.

However, our interest is not with these technical worries that can perhaps be addressed (and that the authors themselves, to some degree, acknowledge). Instead, our worry is with the status of the four basic findings that are presented as good evidence for a representational role on the part of neuronal states. Do these conditions actually qualify as good evidence?

For starters, it is important to not be misled by the author's frequent discussion of the information carried by the neurons, or the use of something with the label "information theory" to describe neuronal activity. Shannon's information theory is a calculus for measuring and formulating degrees of correlative activity between different conditions. Such correlative activity is ubiquitous in most complex input-output systems, including

non-representational ones. In other words, the internal elements of nearly all non-chaotic, non-representational systems carry information and lend themselves to analysis by Shannon's theory. For example, the firing rate of the spark-plugs in a standard combustion engine is correlated with the position of the acceleration pedal. Consequently, the firing rate of the spark-plugs carries information about the pedal's position – the spark plugs' activity is describable using the exact same information-theoretic framing as neurons. But although the spark plugs carry information, they do not *function as* information carriers, (a distinction that will be explained in greater detail below). They do not function as representations. So the fact that information and information theory can be used to describe the degrees to which neuronal activity is correlated with other things does not tell us anything about whether or not they are actually functioning as representations. At best, it tells us that neurons respond to perceptual or other sorts of input in a non-chaotic, non-random manner. In fact, this tendency to describe basic neural causal interactions in informational terms, as "relaying information", "signaling", "feature detecting", "communicating", etc. is ubiquitous throughout the philosophical and neuroscientific literature. Readers should be wary of question-begging descriptions that presuppose a representational role when such a role has not been established.

Turning to the specific characteristics proposed by Pohl, et. al., it seems clear that first three can be combined as the finding that some neurons display the property of being reliably and uniquely responsive to certain distal state of affairs or conditions. The firing rates of the neurons selectively react to the features or states of something else, and only that something else. Their main example captures this – where a neural state is in some way correlated with the presence of a ripe apple, despite variance in, say, lighting or position of the apple. Moreover, Pohl, et. al. are absolutely right to claim that historically this sort of invariant sensitivity has been treated as evidence for neural representations. Going back at least as far as Lettvin et al. (Lettvin et al., 1959) and Hubel and Wiesel (1962), the observation that particular neurons or groups of neurons reliably respond to certain perceptual stimuli – such as the presentation of edges at a certain orientation in the visual field – is regularly treated as strong evidence that these neurons function to encode information. This perspective has been at least partially motivated by the way many conventional representations systems function. A thermometer represents ambient temperature because, in part, the position of the needle or mercury reliably and selectively respond to changes in the ambient temperature. Such devices not only carry information, but are used as information transmitters or presenters. If selective responsiveness is part of what this sort of representation involves, then it is tempting to treat the selective responsiveness of neurons as signifying a representational role.

Still, while such an interpretation might be tempting, we nevertheless must ask if it justified. To what degree does the fact that neuronal states reliably respond to the states of some external condition (or are correlated with the states of some external condition) provide evidence that those neurons are functioning as representations? If we bear in mind the criteria for persuasive evidence discussed in the last section, it seems quite clear that this feature is inadequate evidence for a representational role. The reason is that lots of non-representational roles are carried out by things displaying the exact same sort of selective reactivity. That is, many non-representational systems involve mechanical elements that do their job by going into specific states when, and (for the most part) only when, certain distal conditions obtain. Returning to the firing of the spark-plugs in an ordinary combustion engine, they selectively fire in response to the position of the gas pedal of the car, regardless of the temperature or color of the gas pedal, and they do not respond to the turn signal or the brake pedal. Thus, they exemplify a clearly non-representational role that involves the exact same sort of selective responsiveness. In truth, it is easy to think of many types of non-representational systems, both artefacts and natural systems, in which the internal elements display this sort of distinctive reactiveness. Because selective responsiveness is in no way unique to representations, it cannot serve as serious evidence for representational activity.

Consider this issue with regard to the prime principle of confirmation. Recall that it entails that finding A serves as evidence for hypothesis B over hypothesis C if A is more likely given B than C. Is the selective responsiveness of neurons to specific conditions more likely given the hypothesis that

the neurons are serving as representations, as opposed to the hypothesis that they are playing a different functional role, such as serving as reliable relay circuits? Not at all. One would expect such selective responsiveness to be more or less equally likely with either hypothesis; indeed, with a whole host of possible hypotheses about their functional role. Similarly, with regard to abduction, the hypothesis that a neural structure is functioning as a representation does not best explain the selective responsiveness of the neural structure. That is because that responsiveness can be explained by a number of alternative hypotheses about the neural structure's functional role. Perhaps we should think of these neurons as something more like the "spark plugs of the brain", driving a particular reaction in certain specific conditions without functioning to represent those conditions.[2]

It seems, then, that far from being persuasive evidence for neural representations, the first three findings discussed by Pohl, et. al, are not strong evidence for any such claim. There exist alternative non-representational jobs the neurons could be performing with this sort of responsiveness, such as functioning as stimuli specific relay circuits, or as content-less internal units of a neural network, or as flexible control policies or as other sorts of causal mediators or transducers. Given the way these neural states are triggered, these conditions reveal that neurons do indeed carry information, just like a spark plug. But they do not convincingly reveal that the information they carry is, as such, exploited by the cognitive system.

What about the fourth condition: functionality – the condition requiring the selective activity of the neurons to have a causal influence on subsequent "downstream" processes, eventually contributing to appropriate behavior? Pohl et al. state, "In behavioral tests of functionality, r is functional as a representation of s when r causes s-related behavior b" (2024, p. 16). Here it is the efferent activity that is important, and once again we must ask if this sort of causality provides compelling evidence that neurons are playing a representational role. Unfortunately, the answer here also appears to be "no". Having a beneficial effect that is in some way related to an external triggering condition is not a distinctive feature of any particular functional role, and certainly not unique to a representational role. Pressing the accelerator pedal creates the desired result of the car speeding up, by way of the increased firing of the sparkplugs and their "downstream" influence. But that certainly shouldn't convince us with that the sparkplugs are functioning as representations of the pressed pedal. Indeed, for anything functioning as a reliable causal mediator or condition dependent relay circuit, this sort of causal role is to be expected. If we focus upon human behavior, there are a wide array of biological events that are unique to certain inputs (and carry information about that input) and also contribute to certain behavioral activity, but that are not representations (e.g., smoking causes lung cancer which in turn causes the behavior of quitting smoking, but no one thinks the cancer functions as a representation of smoking). So while functionality is indeed critical, the sort of functionality they describe is not the sort of functionality unique to representational states.

This point can be made once again by considering how the prime principle of confirmation or abduction can guide us in evaluating evidence. As we saw above, it is not the case that the sort of functionality Pohl et. al. describe – generating behavior related to the stimuli – is more likely given a representational role, as opposed to other roles, such as serving as some type of a reliable causal mediator. Consequently, according to the prime principle of confirmation, this functionality is not good evidence for the neural states having a function of representing over other possible functions. Similarly, such functionality is not better explained by assuming the neural state in question is functioning as an information relay. The same behavior could be explained by assuming the neurons are functioning as some sort of causal go-between or non-representational linking liaison

[2] As an anonymous reviewer has pointed out, responsiveness could still be evidence for a representational role over many non-representational roles since unique responsiveness is likely with representation in a way it is not likely with many other roles. For example, unique responsiveness would serve as evidence for a representational role over many sorts of sustaining/regulatory roles that some neurons play, since such a regulatory role would typically not involve unique responsiveness to stimuli. Here again, the value of the evidence would heavily depend upon the explanatory context.

between stimuli and behavior.[3]

While our focus has been on the account provided by Pohl et. al., we believe the difficulties we are raising are quite common and appear in other discussions of evidence for neural representations. For example, Thomson and Piccinini (2018) provide an account of experimental evidence for indicator neural representations that is quite similar. They state, "Observing and manipulating neural representations means establishing, via experimental observations and interventions, that a neural signal fits the criteria for representation. For sensory representations, the criteria are that (1) the signal carries information about some state external to the system, (2) there is a systematic mapping between a range of similar signals and a range of similar external states, and (3) the system uses these internal states to guide behavior" (2018, p. 196).

Here again, we see the property of reliably responding to specific stimuli in a systematic way (with a range of internal states correlating with a range of external states) is treated as evidence for a representational role. And once again, we must point out that such a property would apply to internal states playing various non-representational roles. Like Pohl et al., Thomson and Piccinini also add a downstream condition – that the internal state must guide behavior. But once again, this simply involves creating appropriate behavior given the environmental condition to which the internal state is responding, which is exactly what a good causal mediator does. A good causal mediator reacts to specific input cues in order to generate appropriate effects, given that input. Returning to our favorite example, the sparkplugs (1) carry information about the pedal positions, (2) there is a systematic mapping between a range of sparkplug firings and the range pedal positions and (3) the automobile uses these sparkplugs to guide acceleration behavior. According to Thomson and Piccinini's account of evidence for representations, we also (by extension) have good evidence that automobile sparkplugs are actually functioning as representations. We regard that as a *reductio* of their account of neural representational evidence.

While these accounts focus directly upon the way individual neural states are triggered to support their claim that they are representations, there is another way in which the response profile of neural states is treated as evidence for a representational role. In neural networks it is possible to perform vector analyses of the response profile and receptive fields of internal units, and while the mathematical analyses can become complicated, what is being revealed is not difficult to understand. What has been discovered is that after training, the activation patterns of the system's internal units are not randomly scattered, but instead often form clusters around certain vectors. In other words, some inputs will generate activation patterns in the internal units that cluster in one region of vector space, while other inputs generate activation patterns that cluster in a different region. This reveals that in terms of the internal operations, some types of inputs produce internal states that are similar to one another with regard to their levels of activation, while different types of inputs produce different internal states that are also similar to one another. Here, it isn't simply the fact that neurons respond to specific stimuli that matters. What matters is the similarity in responses in populations of neurons to different (but importantly similar) stimuli. What this finding reveals is that internal elements of the system go into similar sorts of states when the system is presented with similar sorts of input, clustering together in various multi-dimensional mappings of their activities.

For some, this clustering provides evidence that the activity patterns of these internal units are functioning as category representations – as classifying information into conceptual groupings. The claim isn't simply that *if* we assume a representational role for these vectors, we can see that information is categorized in certain ways (Piantadosi et al., 2024). Instead, it is the stronger claim that the clustering in vector space provides

[3] It is also entirely possible that neurons whose activity is correlated with specific stimuli are playing an entirely new sort of role with little or no precedent. Surely we should be open to the possibility that correlative neurons play a novel role we have yet to discover. But such a discovery won't happen if we just assume we already know the role they are playing (and assume that their role is one of representing). Given that the vast majority of biological states and structures are not representations, despite varying degrees of correlation with other conditions, we certainly do not think the default assumption should be a representational role for such neurons.

significant *evidence* for treating these activation patterns of neuron-like units as distributed representations. For example, Shea suggests the vector space clustering itself may be the way we represent the similarity of stimuli – the similarity relation itself is represented by the similarity in vector space (Shea, 2018, pp. 132-134). Or consider this passage from Rupert:

> As a brief illustration, consider subjects who are trained to distinguish two kinds of images, As and Bs, based on the visual features of the stimuli. In doing so, subjects' cognitive or neural systems form "hidden" units (in contrast to input units that detect visual features) the activation of which corresponds systematically to the appearance of relevant stimuli, a1 … an, b1 … bn. Such hidden units cluster into two groups, each of which causes one of two responses, 'A' or 'B' (which then, in a whole agent system, might cause a further action, of, e.g., putting a4 into bin A). The best explanation of how the subject acquired the ability to perform this task function appeals to the correlation between hidden units over which the categorization algorithm operates and the corresponding individuals in the environment. Because that correlation was in place during training, subjects frequently succeeded in the categorization task, which success led to reward and thus to the entrenchment of the categorization algorithm in question. Thus, the units in question represent the corresponding individuals, a1 … an, b1 … bn (Rupert, 2022, pp. 260-261).

Note the reasoning here: clustering in the response profiles, so therefore these structures (or their activation states) are functioning as representations. But how does this clustering stack up as supporting evidence for the existence of representations? Once again, the verdict is not good, and not good because of considerations similar to those described above. Lots of non-representational systems go into a particular range of internal states when presented with one type of input, and a different range of internal states when presented with a different sort of input, and so on. This actually isn't terribly surprising. In Ramsey 2007, this point was made with regard to our digestive system. A vector analysis of the different dimensions of stomach digestion, such as acidity levels, enzyme levels or the rate of peristalsis, would reveal that certain types of food tend to bring about a clustering of states while other types of food would generate different clusters. But the mere fact that there would be these groupings in the response profile is not evidence that these states of the digestive system are functioning as representations; these aspects of the stomach are not playing any such role. They certainly are not functioning to represent the similarity of certain food groups. Congested, urban streets will all tend to create a response profile in spark plug firing patterns that cluster around a certain vector, whereas open highways will create a profile that clusters around a different region of vector space. This fact does not provide us any real reason for thinking that spark plugs are representations, or that their activity profile in a vector analysis is itself a representation. The same point applies to neurons. The observation that neural structures engage in comparable activity when processing related forms of stimuli is fully compatible with those neural structures playing a variety of non-representational roles in the processing. This undermines the claim that such an observation supports a representational hypothesis.

4 Carrying information vs. functioning as an information carrier

The accounts provided by Pohl et. al., Thomson and Piccinini, and various others (see, for example, Shea, 2018) reflect a common way that people think about evidence for neural representations. In the case of Pohl et al., three of the four evidential desiderata pertain to the manner in which the neural state is caused. The first three conditions, sensitivity, specificity, and invariance all describe facets of the process that creates the relevant neuronal activity that is characterized as representational. In Thomson and Piccinini, two of their three conditions are also derived from the casual details of how neural state is generated. This focus upon the triggering etiology of the neural state is standard practice among proponents of neural representations, especially those that treat neural states as reliable indicators. Overwhelmingly, with very few exceptions, philosophers and researchers

direct their attention upon dimensions of how neurons respond to stimuli. The assumption appears to be that evidence for a representational role will mostly depend upon details regarding the generation of the neural state in question.

What motivates this approach? Why do investigators assume that in accounting for neural representations, the overriding emphasis should be focused upon examining and explaining the way neural states are triggered? Normally, when we look for evidence that something is playing a certain role, our attention is directed toward its *performance* and its *impact* on the system. Why instead look at a neural state's responsiveness? After all, being reliably triggered is itself nothing that has any value; such a property is worthless unless it contributes to some more impactful task. Why is there so little discussion of this impact? We can only speculate here, but we believe this oversight likely stems from some combination of two closely related but flawed suppositions: 1) that the possession of content is the critical, defining feature of representation, 2) that possessing content is reducible to possessing information stemming from triggering conditions. Let's consider each in turn.

1) A long-standing tacit assumption in work on cognitive representation has been that the key to understanding representation in the brain is to develop a viable naturalistic theory of content. Content is the special relation between the representation and whatever proposition, property, entity, etc., that is the represented. It is what a functioning representation is "saying". Especially in philosophy, a major industry has been to designate the natural (causal-physical) conditions that "ground" content and that could, it is assumed, thereby make a neural state into an actual representation. With this outlook, content is what converts a neural state into a representation – content is the representation-maker. Consequently, the search for evidence of neural representations has been preoccupied with the search for evidence that neural states possess content (Shea, 2018).

2) Philosophers and researchers often assume that content can be reduced to some complex set of relations and states associated with informational relations. Pohl et. al.'s account is at least in part inspired by so-called informational theories of content presented by people like Fred Dretske, Ruth Millikan and Jerry Fodor. These accounts of content are based upon the observation that things in the world carry information by virtue of being activated in certain conditions. If a neural state is generated when and only when a particular stimulus is present, then that neural state carries the information that the stimulus is present. If it carries the information that the stimulus is present, then (it is presumed) this fact forms the basis for saying the neural state has the content that the stimulus is present, and thus the neural state represents that condition.

In short, it is assumed that content makes something into a representation, and that content is reducible to information carrying arising from the way that something is triggered. However, both of these assumptions are fundamentally flawed. The belief that content is the key to understanding representation gets things exactly backwards. Content is not the representation-maker; instead, representational function is what makes certain natural relations into content relations. Instead of trying to develop a theory of content in order to understand how neurons can function as representations, we should be focused upon developing a theory of representation function to get a better handle on content, and how it comes about through something's playing a representational functional role (Ramsey, 2023).

Moreover, nearly everything in the world carries information in some sense. Carrying information simply means possessing properties which entail facts about other things, and since all objects possess some combination of properties that carry information about certain forces or fields acting upon them, virtually everything is an information carrier. Yet while information-carrying is ubiquitous, relatively few of these things have what we should describe as *content*. To have content, something needs to be a *functioning representation*. We should never think that carrying information about triggering conditions automatically entails the possession of content. Because lots of non-representational things carry information about their causes, it is a fundamental mistake to think that carrying information provides evidence that something is operating as a representation.

By clearing away these misconceptions, we can begin to see that our search for evidence of neural representations has been largely misguided.

We have been trying to understand representation through content and content through information, which has led researchers to obsess over how neural states are triggered. Instead of seeking evidence of neural representation by investigating how they are generated, we should be focusing on details pertaining to what the neural state actually *does*. It bears reiterating that representation is a functional concept or kind (Haugeland, 1991), and something's status as a functional entity depends upon how it acts, not on how it is produced. To qualify as a functionally defined role-player, something needs to bring about the effects associated with that role. If we want to know, say, whether or not something qualifies as a door-stop, it would be a mistake to focus upon how it came about. What we should focus upon is what *it* causes – whether it is used to prevent a door from closing. In the same way, evidence for neural representations ought to be findings or observations that support the claim that neural states are performing the job of representing. Our focus should be upon exploring indications of a certain sort of influence, the sort that of influence that is unique to something's working as a representation.

This carries over directly to the distinction mentioned earlier, between something being an information carrier on the one hand, and something *functioning as* an information carrier on the other hand.[4] This distinction is critical for understanding what qualifies as evidence for neural representations. All sorts of non-representational things (like spark-plugs, heartbeats, tree rings, etc.) carry information, and qualify as information carriers. But most of these things do not function as information carriers. They are not employed *as* information carriers, and thus they do not qualify as representations. Things that carry information can play all sorts of roles, only one of which is the unique representational role. What makes something a representation is how the information it carries is exploited for a specific sort of job; this is what transforms a mere information carrier into a representation (the same point applies to various notions in the same conceptual ballpark as representation, like *detector* or *indicator* or *symbol*).

To help see this point, it is important to recognize that relations that involve information can be exploited in a variety of ways, only some of which exploit the information itself. In other words, the following three conditions can be met: a) something can be recruited and utilized for a functional role because of a relational property it possesses, b) this relational property is such that it carries information about something else, and c) the information that is carried is not germane to the job being performed (even though the underlying relational property is). For example, suppose A is always heavier than B, despite varying weight values for each. The "heavier than" relation can thus be exploited in a variety of ways, only a few of which involve information (and, hence, representation). A *could be* used to relay information about B's maximal weight limit – it could be used to represent the upper bound of possible weight. But A could also be put to work as a counter-weight of B, where the "heavier than" relation is used for a completely nonrepresentational role (Ramsey, 2007, p. 139-140). The upshot of this is that any evidence that a relational property that carries information is exploited by a system is not, as such, evidence that the information itself is being exploited or put to use by the system. Even if we can establish that, through evolution or learning, a neural state comes to play a functional role that involves the exploitation of a stimulus relation that carries information, we really can't use those facts alone as compelling evidence that it is functioning as a representation, since the same relation could be utilized for other purposes.[5]

[4] This is similar to the distinction between something being a shelter and something functioning as a shelter, the latter of which must be the case for something serving as a domicile. A cave might be a shelter, but if there is no evidence that anything has ever used it as such, then there is no evidence that it ever functions as a shelter and thus no evidence it qualifies as a domicile.

[5] A reviewer has noted that someone might try to justify interpreting neurons as representations because the brain is commonly viewed as an information-processing system; the information-processing perspective could then perhaps be used to license interpreting neural activity in representational ways. However, such an interpretation would get the conceptual dependency backwards. The information-processing paradigm is actually based upon the conceptually prior assumption that the system uses representations. We can't justify invoking representations by appealing to something that is itself entirely dependent upon the presupposition that representations exist.

5 Efferent evidence?

If evidence supporting a representational interpretation of neural states should be based upon what the neural state is actually *doing*, then our search for such evidence should be focused upon the efferent, downstream effects and influences of the state. To adopt the terminology of Millikan (1984) and other teleosemanticists, what should matter is how the neural state's activities, or the information it carries, is "consumed". Aren't there some accounts that use the causal consequences of neural state activity as evidence for neural representations?

We have already seen in Section 3 above that one type of efferent activity that is treated as evidence for a representational role is the generation of useful or adaptive motor output that is appropriate *in light of* the presence of the allegedly represented stimuli. If a neural state responds to all and only predators, and if this causes the neural state to produce fleeing behavior, then this behavior is thought to support interpreting the state as playing a representational role (Dretske, 1988). However, as we saw, serving as such a mediator between specific environmental conditions and suitable behavior is not a role unique to representations. Lots of types of adaptive relay circuits are going to have the same basic causal profile. Such a role can certainly serve as good evidence that the neural state is important and beneficial. But it is not good evidence that it is functioning to represent the relevant environmental condition, any more than adrenalin production is functioning to represent perceived danger. Remember, evidence that a relational property is critical for a role is not the same thing as evidence that any information associated with that relational property is critical for that role.

Another common strategy is to offer a semantic interpretation of the efferent effects themselves, treating them as a form of prescriptive content. If we just assume the neural state is representing an imperative instruction, then it its effects will be characterized as what is represented. For example, Thomson and Piccinini invoke findings that they propose support a representational role for motor neurons, which they claim "imperatively represent" commands for certain motor output. They state, ". . . a state S imperatively represents that P just in case the system that produces S has the function of producing S in order to bring it about that P. That is the system's imperative representational function" (2018, p. 196).

The problem with this way of understanding evidence for neural representation is that it doesn't actually involve any sort of discovery or evidence that something is functioning as a representation. Instead, it involves writers simply assuming or stipulating an imperative representational role, and then announcing that the neural states' effects qualify as a type of content. This approach could be used to describe all things that have the job of causing something else to happen in specific conditions. Sparkplugs have the job of generating explosive piston motion (which in turn causes increased drive-shaft rotation), increasing when the acceleration pedal is pressed. If we just presuppose or stipulate a representational role, we could then describe the sparkplugs as imperatively representing the command "Piston, move away now!!". But this is the sort of invoking of representational language that has no explanatory pay-off. Condition-dependent causality is no more likely and no more easily explained by assuming a representational role than it is by assuming other mechanistic roles. If we are interested in what should count as a good evidence for neural representations, this approach is a non-starter.

Perhaps one of the most sophisticated attempts to explain what operating as an internal representation involves has recently been provided by Nicholas Shea (2018). Shea offers a detailed account of how certain head-world relations, like correlations or structural similarity, can be exploited to give rise to functioning cognitive representations because of efferent activity. Before looking at his account of evidence for neural representations, a couple of noteworthy features of Shea's account need highlighting. First, a great deal of Shea's narrative involves what he describes as "how possible" case studies. That is, much of his discussion involves possible processes and conditions that may or may not be actually be implemented in neural structures. Consequently, he is less focused upon providing actual evidence of neural representations and more focused upon ways in which representation *could* happen in the brain, using scientific theories serving as *possible* applications of his ideas. The second point is that al-

though Shea does indeed discuss neural states in representational roles, his most plausible proposals involve the implementation of computational algorithms in which representational vehicles are viewed as computational states. For example, in many of his proposals, representational vehicles are unspecified structures that represent numerical values in the execution of a mathematical computation. Recall that our issue is not with the existence of cognitive representations at higher levels of analysis, such as what might be involved in the execution of computational processes. Insofar as Shea's often intriguing speculations are less about neural activity, as such, and more about higher level, algorithmic cognitive operations, his proposals are not the target of our primary concerns.

Nonetheless, Shea's account does attempt to provide what he calls an "evidential test" for when representational information is exploited by a system and thus for when something, including neural states, function as a representation (Shea, 2018, pp. 89-91; 142-143). The evidential test focuses upon what happens to performance when there is some sort of breakdown in the informational relation, say, by the introduction to noise into the system that undermines the degree of correlation or structural similarity between neural states and their targets. If the degradation of the relation substantially undermines the practical value of the behavior, Shea suggests, then that is evidence that the information is what is being exploited and that the state in question is a representation.

However, to repeat a common theme, the phenomenon that Shea picks out – less adaptive behavior resulting from some sort of breakdown in head-world relations – is a phenomenon that also pertains to non-informational uses of relational properties and non-representational roles of neurons. Insofar as the system exploits any relational properties of an internal state or structure for successful performance, that performance will always be degraded when the relation itself is degraded. That fact alone doesn't tell us *how* the relational properties are exploited, or that they involve the exploitation of information. If there is some sort of a breakdown in the causal connection between the car's gas pedal and spark plug activity, then the performance of the vehicle will suffer. That fact doesn't entail the spark plugs are functioning as information carriers. Put more generally, we can't use evidence of some sort of malfunctioning as evidence for flawed representation. Improper functioning alone never amounts to misrepresentation unless we have already established that the thing malfunctioning is indeed functioning as a representation. Indeed, non-representational control policies operating on internal states will malfunction in exactly the same way (see Krakauer, 2019).

What about the discovery that there exists a degree of structural similarity or homomorphism between neural structures on the one hand, and some external state of affairs on the other hand? Could this serve as evidence for a certain sort of mapping or modeling role? It is tempting to think so because external maps, models and simulations carry information not by being reliably triggered by certain conditions, but by (roughly) sharing, in some way, a structure with those things and events they represent. We have already noted how Shea suggests that proximity of hippocampal place cells in vector space may function in a map-like way, representing actual proximity of environmental location. And Thomson and Piccinini suggest that the discovery of a "topographic map of visual space" in the V1 region of the visual cortex serves as evidence that these neurons in V1 are representations (2018, p. 12-14).

As we noted above, we are not opposed to the claim that cognitive systems use cognitive maps or models that are somehow, perhaps computationally, implemented in the brain. We believe there is indeed evidence that such structures exist and function as representational systems at some higher level of analysis. Our concern here is whether or not there exists evidence that such systems exist at the level of neuronal interactions – whether or not there is justification for thinking action potentials and synaptic interfaces are themselves mechanisms that should be viewed as representational maps, models or simulations. Even if such neural structures carry information through some sort of structural similarity with something else, is there evidence that they are actually *functioning as* information carriers? Our primary concern is with the inference that goes from the observation of structural homomorphism between neural mechanisms and stimuli (and thus the carrying of structural information about that stimuli) to the conclusion of a representational role for the neural

mechanism. Evidence for the former is not compelling evidence for the latter because, to repeat a common theme, various non-representational roles might give rise to such homomorphisms. To return to our favorite analogy, there exists a structural correspondence between the range of spatial positions of the accelerator pedal and the array of firing rates of the sparkplugs. There is a clear homomorphism between the two realms. However, this clearly shouldn't lead us to think the sparkplugs firings are functioning to model the activity of the accelerator pedal. What is needed is evidence that the structural information itself is actually being exploited *as* information about the properties of something else. The existence of some form of structural similarity between neural states and external conditions merely implies the neural states have the *potential* to function as structural representations, not that they actually do.

In a recent publication, Marco Facchin (2023) has provided a much more detailed critique of the evidence for structural representations at the neural level of analysis. Facchin examines various claims used to support the discovery of allegedly structural neural representation vehicles and argues that none of these withstand scrutiny. We largely agree with Facchin's analysis; however, we would add to Facchin's critique that for many alleged structural neural representations, such as the place cells in hippocampal "maps", neural states are *not* treated as representations because they are functioning as surrogates or stand-ins in the way typical map constituents function. Unlike the markings on a map, a given neural state's relations to other neural elements is not treated as a motivation for the representational interpretation. Instead, they are treated as representations because they reliably respond to specific stimuli, just as we saw with standard neural representations. In other words, despite talk of a map-like format for these representations, at the end of day the place cells are treated as place representations primarily because they reliably respond to the proximity of places, and not because they participate in some broader map or model. Thus, the problems that we've discussed regarding using responsiveness as evidence for representational roles extend to these accounts.

Finally, what about the *lack of* efferent activity as evidence for representation? Some writers have suggested that "off-line" processing, in which internal activity is disconnected from overt behavior can serve as evidence that the internal activity is representational in nature (see, for example, Shea, 2018, pp. 115-116). This outlook is motivated by the fact that preparation, recitation and counter-factual reasoning typically involves the employment of representational models which are exploited for rehearsal and pre-planning. When neural activity associated with specific output takes place without such output, this is regarded as an indication that in this context, such a model is in use for planning/rehearsal purposes.

The problem with this outlook is that to interpret the activity as a sort of pre-planning or counterfactual modeling, one must already assume that the activity in question is representational in nature. Merely observing neural activity disconnected from behavior doesn't really tell us anything about the functional role of the activity, and is not stand-alone evidence for neural states playing a representational role. In fact, if anything, such an observation would serve as *disconfirming* evidence for any hypothesis claiming that the activity is functionally connected to the behavior. It would serve as evidence that there may be alternative interpretations of the activity since it implies the activity is not designed to generate the behavior in question.

But let's suppose we can be quite confident that in its normal mode of operation, the neural activity is indeed supposed to generate the efferent output that is absent in an alternative mode of operation. Thus, this alternative mode of operation involves a type of off-line activation that happens, let's assume, for some alternative purpose. Must that purpose be the modeling of some environment or action, and thus reveal the employment of representations? It does not. We know that neural connections are reinforced and strengthened through repeated activity. Suppose the neural circuitry required for some action is quite complicated and would benefit from long term potentiation through synaptic augmentation via repeated engagement. The benefit of such off-line activity would apply to all sorts of functional roles, including, say, various non-representational flexible control policies or neural circuits engaged in something other than modeling or simulation. If there is value in the offline reinforcement of neural connections, and if that value pertains to non-representational func-

tional roles that neural states could be playing, then we cannot use offline reinforcement as evidence for a representational role. Once again, we can't treat something as evidence for neural representation if the finding can be expected to accompany a variety of nonrepresentational roles.

6 What should we be looking for?

So far our analysis has been critical. We have been ruling out various findings as contenders for serving as noteworthy evidence that neural states are functioning as representations. But it is fair to ask what *could* reveal that an internal structure of some kind is functioning as an information carrier, and thereby as a representation, and not functioning in some other role? What should we be looking for as actual evidence that internal states are representations?

While this is a complex topic that deserves its own article-length treatment (see Ramsey, forthcoming), we can here touch on at least one sort of consideration that might support a representational interpretation. Above it was suggested that because there is no inner homunculi "interpreter" or "informee" reading off the meaning of inner representational states, we must try to make sense of representation function in a way that is substantially different from how we use things like thermometers or maps. But perhaps this was too quick. Perhaps it is indeed possible to try to understand how representation consumption can occur in the brain in a manner that more closely resembles our normal use of representational entities. We already know how computational systems can perform cognitive operations like inference, interpretation, translation, etc. Instead of simply pretending that any sort of helpful downstream processing qualifies as representation consumption, we should search for efferent activity that actually involves the sorts of things representation use commonly entails.

For example, if an investigator hypothesizes that a neural network is somehow functioning as an internal map, then search for evidence of computational operations that are sensitive to, take advantage of, and make inferences based upon the relational properties between the component representational elements of the purported neural map. That would provide real evidence that the neural structure is functioning as a structural representation because it would involve operations similar to the ones we perform when using structural representations. We examine the structural relations of the token components of a map to make inferences about real world entities and their relations to one another. This is what Swoyer (1991) referred to as *surrogative reasoning*, whereby structural relations between representational elements are used to make analogical determinations about real world conditions and relations. If there is evidence that such computational operations are taking place, then that could possibly provide evidence that something is functioning as a structural representation.

A traditional worry for this approach has been the concern that the invoking of a robust representation-user would lead to a regress, since the representation-user would need its own internal representations to perform the sophisticated activity of interpreting inner representations, and it is assumed these new internal representations in the representation-user would require their *own* little inner-users, and so on and on. But we regard this as a pseudo-problem. There are various ways in which cognitive sub-systems can take advantage of representational states in the broader system without themselves employing their own internal representations. For example, it is possible for the sub-system's functional or computational architecture to embody the kind of sophisticated aptitude needed to employ other states as representations, but to do so in a way that involves no further representational states inside the sub-system.[6] The key point is that it is possible to make sense of cognitive processes that engage in

[6] Dennett (1983) and Clapan (2002) have argued that one way in which a representational regress can be avoided is if the representation user does not employ explicit, discreet representations, but instead performs operations through the use of some sort of tacit or implicit representation. However, one of us (Ramsey, 2007, 2023) has argued that there is no such thing as tacit representation (because representation requires some actual item or entity playing a representational functional role); consequently, the proper way to make sense of tacit knowledge is to suppose that it is a sort of aptitude or capacity that does not rely on any sort of representations. Such an analysis makes it even easier to see how modules could function as representation-interpreters without any threat of a regress.

sophisticated operations and that treat states as representations, but without such a treatment leading to any sort of representational regress (Ramsey, forthcoming).

In fact, we already have an understanding of cognitive operations that actually do, in a sense, treat their inputs as representations because they engage in computations. Consider a mathematical operation such as division. Division, as such, involves the conversion of numbers (dividend and divisor) into a different number (the quotient). However, computational systems cannot convert non-physical abstract entities like numbers, so instead they convert entities that represent numbers. In other words, to make sense of a physical system like the brain engaging in this sort of computation, we must regard them as treating their input states as representations of other things (Ramsey, 2007). This is certainly true of mathematical operations, but it is also true of many other sorts of computational processes that are commonly regarded as crucial for cognition. In these cases, what makes the state in question a functional representation is not any head-world relation that carries information. What makes the state in question a representation is how it is actually utilized by a module engaging in computations. Evidence for such computational operations taking place in the brain would qualify as evidence that the system is utilizing this sort of input-output representation.

However, as evidence for *neural* representation, there is a major problem with this kind of evidence. For this type of representational activity, the level of description would involve fairly high-level computational operations. The sort strategic inferences we make when using structural representations like maps are fairly sophisticated; if something similar occurs in the use of internal representations, it would require computational operations that are more algorithmically defined than operations described at the level of neuroscience. The same is true for the sort of computational operations like mathematical calculations. As we noted at the outset, we are not opposed to cognitive representations involving computational models or more abstract algorithms. We are not denying that, at higher and more abstract levels of theorizing about cognitive processes, representational theorizing has legitimacy. Our doubts are about the invoking of representations at the level of neurons and action potentials.

It might be protested that computations and algorithms are indeed within the explanatory domain of neuroscience, especially the field of cognitive or computational neuroscience. In response, we certainly do not want to insist upon overly restrictive boundaries on what does or doesn't count as neuroscience. Moreover, we acknowledge that distinguishing different level of description or analysis is itself a difficult and contentious topic. But as we noted above, what matters is how the explanatory constructs and processes are characterized. If they are presented in purely computational terms, such that they could be instantiated by something other than actual neural states and structures, then we have a hard time understanding the claim that there is evidence for neural structures, as such, functioning as representations. If the theoretical level of description does not actually involve neurons or arrays of neural structures, and more closely resembles the posits and computations associated with a standard cognitive psychological theory, then it is far from clear why these should be regarded as *neural* representations.[7]

7 Conclusion

There are, no doubt, various ways in which researchers and theorists have invoked evidence for neural representations that we have not been able to discuss here. Are goal has been to simply expose the inadequacies of some of the more popular forms of alleged evidence for neural representation, and to provide a framework for properly evaluating any evidence source. It is crucial to recognize that the quality of representational evidence is always dependent upon the possibility of alternative, non-representational roles for

[7] One possible way in which we one could have neural representations even if higher level computations were involved would be the following: while the computations themselves would be higher level, they would be responding to the actual physical nature of (or relations between) neural states, in a sense interpreting them as something like a map or model. This could perhaps be a sort of hybrid picture in which the high-level computations would be about neural states and their relations to one another.

neural states that the finding could also support. Thus, investigators should always ask themselves if the representational interpretation of the data is warranted, or if instead there are other possible (non-representational) interpretations that fit the data. Moreover, if investigators and philosophers seek evidence for neural representations, then they should be focused upon details pertaining to what the neural state in question actually does, and not how it is triggered – on how the system responds to it, not on how it responds to something else.

Recently, it has been demonstrated just how imprecise the concept of representation is in cognitive science – including inside of sub-disciplines like neuroscience – with different investigators apparently adopting substantially different understandings and criteria (Favela and Machery, 2023). Among the many ways this imprecision will harm theoretical progress, foremost is with regard to our understanding of what counts as empirical evidence for the presence of representations. Our aim has been to convince you that what is often presented as evidence for neural representations actually isn't, and that researchers need to more carefully consider other possible interpretations of their findings and other possible non-representational roles for neural states. At the present, there really does appear to be something like a "representation-attribution-error" that is pervasive in the neurosciences. While clarification, reform and "precisification" of the concept of representation will substantially improve our understanding of what should and shouldn't count as evidence, it is also possible that enhanced understanding can move in the opposite direction. By thinking more carefully about suitable evidence for representations, and about how various findings are consistent with non-representational roles, researchers will gain a better understanding of the notion of representation itself, the sort of explanatory roles it should be playing in various accounts, and why it may not actually apply to neural states as frequently as alleged.

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